

Exponential functions



- 1 a 5 b 80 c $\frac{5}{4}$ d 320
- 2 a 1 b $\frac{9}{16}$
- 3 50 313.28
- 4 a 125 b 1 c $\frac{1}{5}$ d -2
- 5 a (5, -32) b $\left(-3, -\frac{1}{8}\right)$ c $\left(-1, -\frac{1}{2}\right)$ d (4, -16)
- 6 $m = 8$
- 7 a $n = 8$ b $k = 4$ c $m = -2$
-

Characteristics of exponential functions

- 8 No, both functions are always greater than zero.
- 9 None, for all values of x it will have a positive value.
- 10 a (0, 1) b (0, 1) c (0, -1) d (0, -1)
- 11 a 1 b 1

12

a $\frac{1}{81}$

c 81

e It gets smaller, approaching zero.



b 1

d It gets larger, approaching infinity.

13

a Positive

b 0

c $y = 0$ is a horizontal asymptote of the curve.14 There is no maximum function value. As x decreases, y approaches 0 but never quite gets there.

15

a $y = 4^x$

b y is increasing at an increasing rate.

16 $y = 5^x$

17 $y = 5^{-x}$

18

a

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y	$\frac{1}{243}$	$\frac{1}{81}$	$\frac{1}{27}$	$\frac{1}{9}$	$\frac{1}{3}$	1	3	9	27	81	243	59 049

b As x increases, the function increases at a faster and faster rate.c All real x

d $y > 0$

19



a

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y	32	16	8	4	2	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{1024}$

b As x increases, the function decreases at a slower and slower rate.

c 1

d All real x e $y > 0$

20

a

x	-5	-4	-3	-2	-1	0	1	2	3	4	5	10
y	$-\frac{1}{32}$	$-\frac{1}{16}$	$-\frac{1}{8}$	$-\frac{1}{4}$	$-\frac{1}{2}$	-1	-2	-4	-8	-16	-32	-1024

b No. For $a > 0$ the expression a^x is greater than zero for all real x . Now, $-(a^x)$ will be less than or equal to zero.

c Decreasing

d As x increases, the function decreases at a faster and faster rate.e All real x f $y < 0$

21

a $(0, 1)$

b No

c All real x d $y > 0$ e $y = 128$ f $x = 8$

22

a Example answers:

- None of the curves cross the x -axis.
- They all have the same y -intercept.
- As x gets large, the function values approach 0.

b 1

c As x gets large, the function values approach 0.

23

a $(0, 1)$ b As x gets large $y = 3^x$ becomes large while $y = 3^{-x}$ approaches 0.

24

a No

b Yes

c Yes

d No

Graphs of exponential functions

25

a Yes

b No

c No

d No

e No

f No

g No

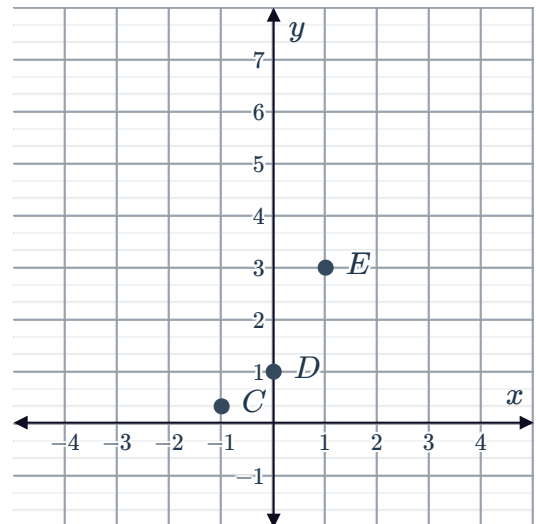
h Yes

26

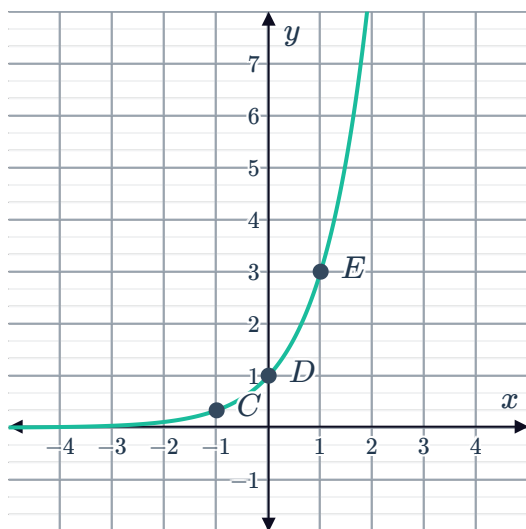
a

Point	Coordinates
A	$\left(-3, \frac{1}{27}\right)$
B	$\left(-2, \frac{1}{9}\right)$
C	$\left(-1, \frac{1}{3}\right)$
D	$(0, 1)$
E	$(1, 3)$
F	$(2, 9)$
G	$(3, 27)$

b



c

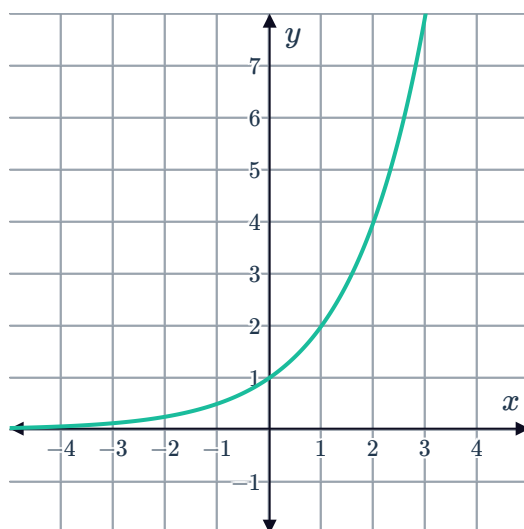


27

a

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	8

b



c Increases

d Decreases

e The curve never crosses the x -axis.

f 1

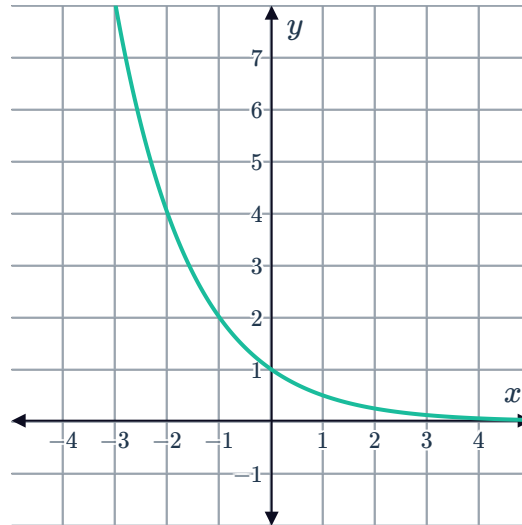
28



a

x	-3	-2	-1	0	1	2	3
y	8	4	2	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$

b



c Decreases

d Increases

e The curve never crosses the x -axis.

f 1

29



a

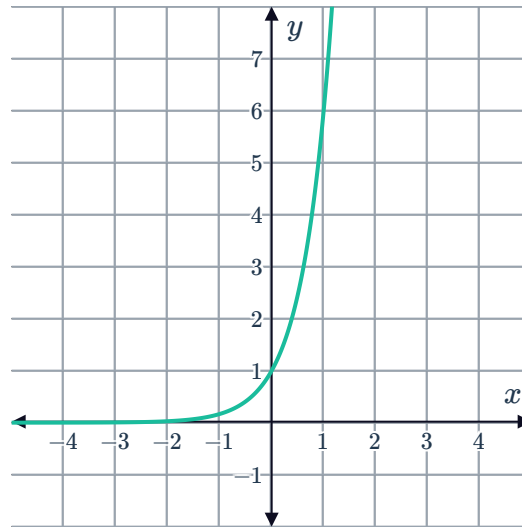
i (0,1)

ii

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{216}$	$\frac{1}{36}$	$\frac{1}{6}$	1	6	36	216

iii $y = 0$

iv



b

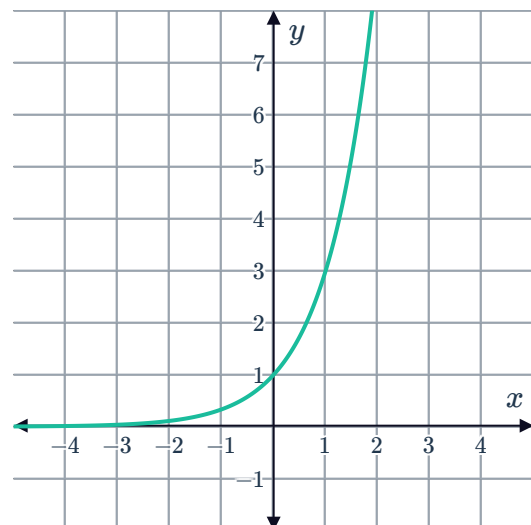
i (0,1)

ii

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{27}$	$\frac{1}{9}$	$\frac{1}{3}$	1	3	9	27

iii $y = 0$

iv



c

i $(0,1)$

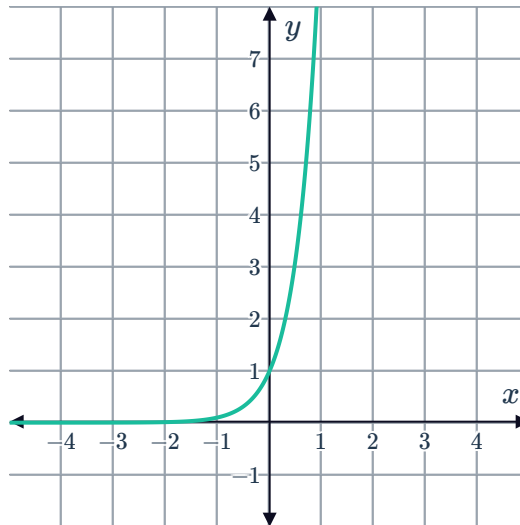


ii

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{1000}$	$\frac{1}{100}$	$\frac{1}{10}$	1	10	100	1000

iii $y = 0$

iv



d

i $(0,1)$

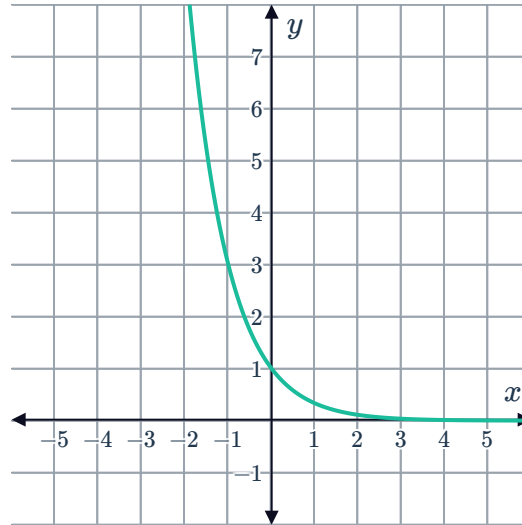


ii

x	-3	-2	-1	0	1	2	3
y	27	9	3	1	$\frac{1}{3}$	$\frac{1}{9}$	$\frac{1}{27}$

iii $y = 0$

iv



30

a No. For $a > 0$ the expression a^x is greater than zero for all real x .

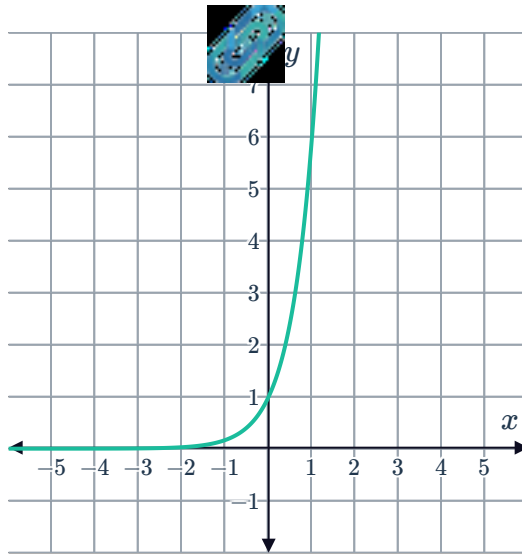
b ∞

c 0

d 1

e 0

f



31

a No. For $a > 0$ the expression a^x is greater than zero for all real x .

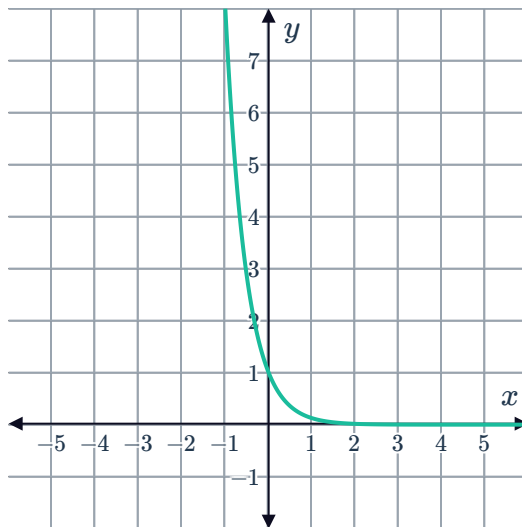
b 0

c ∞

d 1

e 0

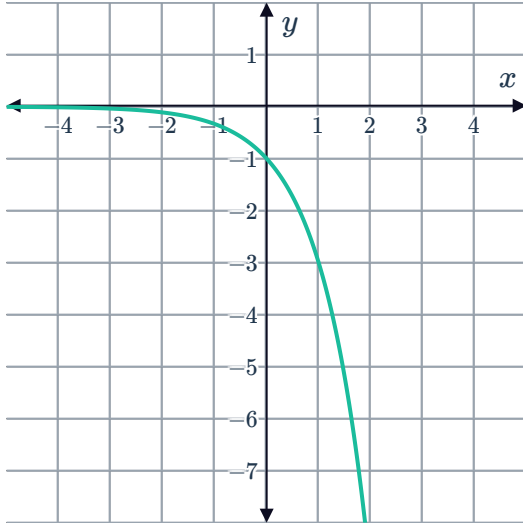
f



32

a $(0, -1)$

c



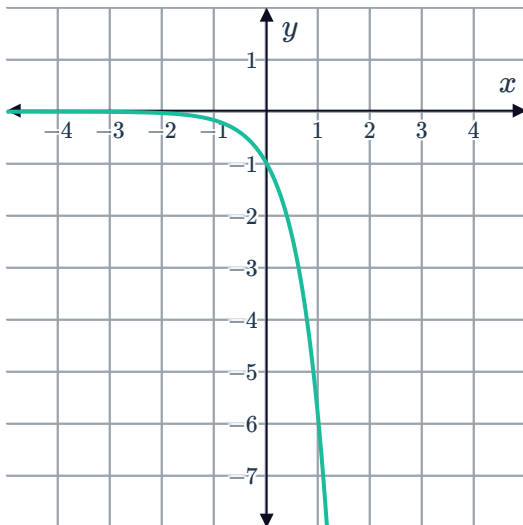
b $y = 0$

d Decreasing function

33

a $(0, -1)$

c



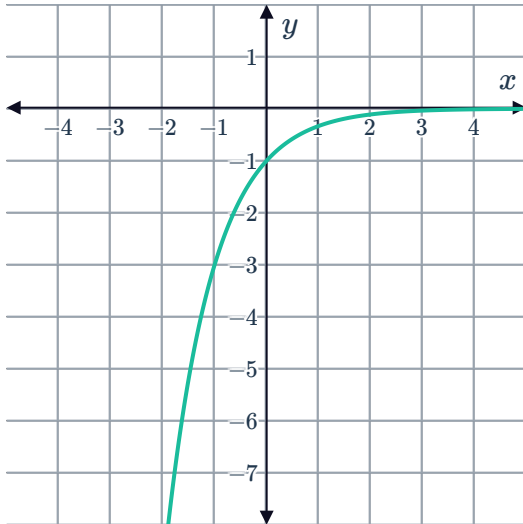
b $y = 0$

d Decreasing function

34

a $(0, -1)$

c



b $y = 0$

d Increasing function

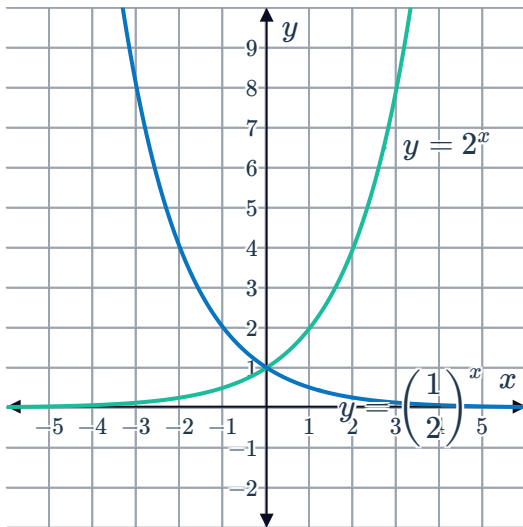
35

a

i Yes ii Yes iii No iv No

b Reflection across the y -axis.

c

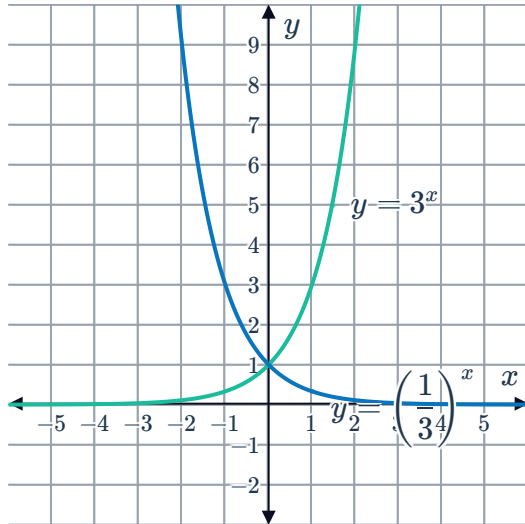


36

a $y = 3^{-x}$

b Reflection about the y -axis.

c



37

a $y = 0$

b $y = 0$

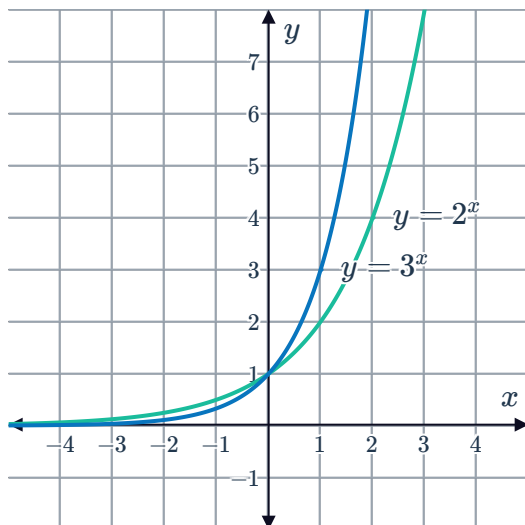
c

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{27}$	$\frac{1}{9}$	$\frac{1}{3}$	1	3	9	27

d

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	8

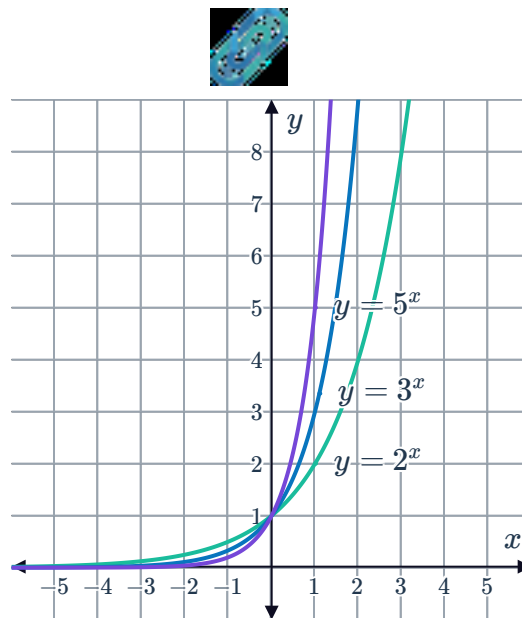
e



f (0, 1)

38

a



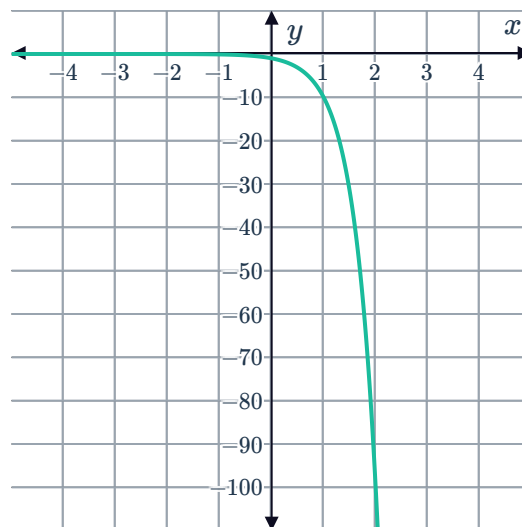
- b Shared y -intercept of (0, 1)
- c As x gets larger, the function increases at an increasing rate and the values approach ∞ .

39

- a Graph R
- b For $x < 0$, the graph of $y = 6^x$ is below the graph of $y = 4^x$. This is because negative values of x result in fractional function values, and for any negative value of x , 6^x will result in a smaller value than 4^x .

40

- a Yes. For any value of x , 10^x is always positive. Therefore, -10^x is always negative.
- b



- c No solution